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Novel and efficient FDTD implementation of higher-order perfectly matched layer based on ADE method

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ABSTRACT

Based on the stretched coordinate perfectly matched layer (SC-PML) formulations and the auxiliary differential equation (ADE) method, an efficient and unsplit-field implementation of the higher-order PML scheme with more than one pole is proposed to truncate the finite-difference time-domain (FDTD) lattices. The higher-order PML has the advantages of both the conventional PML and the complex frequency shifted PML (CFS-PML) in terms of absorbing performances. The proposed algorithm is validated through two numerical tests carried out in three dimensional and two dimensional domains. It is shown in the numerical simulations that the proposed PML formulations with the higher-order scheme are efficient in terms of attenuating both the low-frequency propagating waves and evanescent waves and reducing late-time reflections, and also hold much better absorbing performances than the conventional SC-PML and the convolutional PML (CPML) with the CFS scheme.

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1. Introduction

Since the introduction of the perfectly matched layer (PML) absorbing boundary condition by Bérenger [1], various modified PMLs have been presented to terminate the finite-difference time-domain (FDTD) lattices. With the advantage of simple implementation in the corners and the edges of the PML regions, the stretched coordinate PML (SC-PML) [2] was proposed through mapping Maxwell's equations into a complex stretched coordinate space. As original Bérenger's PML, the SC-PML formulations in [2] are ineffective at absorbing the evanescent waves. Recently, the complex frequency shifted PML (CFS-PML) [3], implemented by simply shifting the frequency dependent pole off the real axis and into the negative-imaginary half of the complex plane, has drawn considerable attention due to the fact that this PML is efficient in attenuating the low-frequency evanescent waves and reducing late-time reflections [4]. In [4], the convolutional PML (CPML), based on the SC-PML formulations and the convolution theorem, was presented in detail to efficiently implement the CFS-PML. However, the CFS-PML would have a poor absorption of low-frequency propagating waves as shown in [5–7]. In [7], the performance of regular PML, complex frequency shifted PML, and second-order PML is studied for the numerical stimulation of waveguide problems. The limitation of each PML is clearly demonstrated. The regular PML is the best choice when only propagating waves are present. The CFS-PML is the best choice if strong evanescent waves are present but low-propagating waves are absent. In more general case, where both low-propagating waves and strong evanescent waves are present, the second-order PML is the best choice. It is shown in the numerical stimulations that the second-order PML can attenuate both low-propagating waves and strong evanescent waves. To overcome the limitations of both the conventional PML and the CFS-PML, the higher-order PML was proposed by Correia, which retains the advantages of both the CFS-PML and

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conventional PML in [7]. It has shown that the second-order PML is highly effective in absorbing both evanescent and low-frequency propagating waves in both open-region and periodic problems in [8]. In [8], the concept of a higher-order PML is proposed, and a split-field implementation of higher-order PML is proposed based on SC-PML and ADE method to truncate FDTD lattices. The proposed higher-order PML in [8] has the advantages of both the conventional PML and the complex frequency shifted PML in terms of absorbing performances. However, besides the drawback of more requirements of the memory and the computational time, the higher-order PML implementation proposed in [8] was difficult to be extended to the case with more than two poles because the polynomial expansion was employed.

In this paper, an efficient implementation of the higher-order PML based on unsplit-field SC-PML formulations and the auxiliary differential equation (ADE) method is proposed. For convenience, this PML is referred to here as the LYYADE PML. Consequently, the proposed higher-order PML scheme requires less memory and computational time as compared with that in [8]. Only the second-order case is described in this paper, but this approach is easy to be applied to any number of poles. Due to using unsplit-field, the proposed formulations can save more auxiliary variables than [8]. Besides, due to applying D-B formulations, referring to [12], Consequently, this PML can be applied to truncate arbitrary media, such as lossy, dispersive, anisotropic, inhomogeneous or nonlinear without any modification and all that is needed is to modify (8) and (9) under consideration. The method is available in [12] to obtain E from D using (8) [and H from B using (9)]. It must be noted that if $\varepsilon_r(\omega)$ [or $\mu_r(\omega)$] are not frequency-dependent, E (or H) formulation should be adopted to reduce memory requirement and save computational time. Finally, in [8], the second-order PML based on the SC-PML was implemented by using the split-field PML formulations. However, besides the drawback of more requirements of the memory and the computational time, the higher-order PML implementation proposed in [8] was difficult to be extended to the case with more than two poles because the polynomial expansion was employed.

2. Formulation

In three-dimensional (3-D) SC-PML regions, the normalized frequency-domain modified Maxwell's curl equations can be written as

$$j\omega\varepsilon_r(\omega)\mathrm{E}(\omega) = c_0\nabla_s \times \mathrm{H}(\omega) \tag{1}$$

$$j\omega\mu_r(\omega)\mathrm{H}(\omega) = -c_0\nabla_s \times \mathrm{E}(\omega) \tag{2}$$

where c_0 is the speed of light in free space, $\varepsilon_r(\omega)$ and $\mu_r(\omega)$ are, respectively, the relative permittivity and permeability of the FDTD computational domain and the operator ∇_s is expressed as

$$\nabla_s = \hat{x}S_x^{-1}\partial_x + \hat{y}S_y^{-1}\partial_y + \hat{z}S_z^{-1}\partial_z \tag{3}$$

where ∂_x, ∂_y and ∂_z are the partial derivatives with respect to x, y and z and S_η ($\eta = x, y, z$) are the complex stretched coordinate metrics, which was originally proposed [1] to be

$$S_\eta = 1 + \sigma_\eta/j\omega\varepsilon_0, \quad (\eta = x, y, z) \tag{4}$$

with the CFS scheme and S_η ($\eta = x, y, z$) are defined as

$$S_\eta = \kappa_\eta + \sigma_\eta/(\alpha_\eta + j\omega\varepsilon_0) \tag{5}$$

where α_η and σ_η are assumed to be positive real and κ_η is real and $\geqslant 1$. To make the PML completely independent of the material properties of the FDTD computational domain, both (1) and (2) can be written in terms of the electric flux density D and the magnetic flux density B as

$$j\omega\mathrm{D}(\omega) = c_0\nabla_s \times (\omega) \tag{6}$$

$$j\omega\mathrm{B}(\omega) = -c_0\nabla_s \times (\omega) \tag{7}$$

where D and B are given by

$$\mathrm{D}(\omega) = \varepsilon_r(\omega)\mathrm{E}(\omega) \tag{8}$$

$$\mathrm{B}(\omega) = \mu_r(\omega)\mathrm{H}(\omega) \tag{9}$$

Consequently, this PML can be applied to truncate arbitrary media, such as lossy, dispersive, anisotropic, inhomogeneous or nonlinear without any modification and all that is needed is to modify (8) and (9) under consideration. The method is available in [12] to obtain E from D using (8) [and H from B using (9)]. It must be noted that if $\varepsilon_r(\omega)$ [or $\mu_r(\omega)$] are not frequency-dependent, E (or H) formulation should be adopted to reduce memory requirement and save computational time.The idea of the higher-order PML was proposed in [8] by generalizing this metric for the case where more than one pole was present. For the second-order PML, S_η is defined as

$$S_\eta = S_{1\eta} \cdot S_{2\eta} = (\kappa_{1\eta} + \sigma_{1\eta}/(\alpha_{1\eta} + j\omega\varepsilon_0)) \cdot (\kappa_{2\eta} + \sigma_{2\eta}/(\alpha_{2\eta} + j\omega\varepsilon_0)) \tag{10}$$

As an example, consider the discretization of the x component of (6) in all corners and the edges of the PML regions which are run parallel with the x direction (i.e. $S_\eta = \kappa_\eta + \sigma_\eta/(\alpha_\eta + j\omega\varepsilon_0)$, $(\eta = y, z)$, are chosen.):

$$j\omega D_x = S_y^{-1}\partial_y H_z - S_z^{-1}\partial_z H_y \tag{11}$$

Using the partial fraction expansion, S_η^{-1}, $(\eta = y, z)$ can be written as

$$S_\eta^{-1} = S_{1\eta}^{-1}\cdot S_{2\eta}^{-1} = (\kappa_{1\eta} + \sigma_{1\eta}/(\alpha_{1\eta} + j\omega\varepsilon_0))^{-1}\cdot(\kappa_{2\eta} + \sigma_{2\eta}/(\alpha_{2\eta} + j\omega\varepsilon_0))^{-1} \tag{12}$$

And arranging, S_η^{-1} can be rewritten as

$$S_\eta^{-1} = (1/\kappa_{1\eta}\kappa_{2\eta})(\kappa_{1\eta}(\alpha_{1\eta} + j\omega\varepsilon_0)/(\kappa_{1\eta}(\alpha_{1\eta} + j\omega\varepsilon_0) + \sigma_{1\eta}))(\kappa_{2\eta}(\alpha_{2\eta} + j\omega\varepsilon_0)/(\kappa_{2\eta}(\alpha_{2\eta} + j\omega\varepsilon_0) + \sigma_{2\eta})) \tag{13}$$

Numerator and denominator multiply by $(\kappa_{1\eta}\varepsilon_0)^{-1}\cdot(\kappa_{2\eta}\varepsilon_0)^{-1}$, respectively, we can obtain

$$S_\eta^{-1} = (1/\kappa_{1\eta}\kappa_{2\eta})((\alpha_{1\eta}/\varepsilon_0 + j\omega)/(\alpha_{1\eta}/\varepsilon_0 + \sigma_{1\eta}/\kappa_{1\eta}\varepsilon_0 + j\omega))((\alpha_{2\eta}/\varepsilon_0 + j\omega)/(\alpha_{2\eta}/\varepsilon_0 + \sigma_{2\eta}/\kappa_{2\eta}\varepsilon_0 + j\omega)) \tag{14}$$

And rearranging, S_η^{-1} can be rewritten as

$$1/S_\eta = c_\eta((a_{1\eta} + j\omega)/(b_{1\eta} + j\omega))((a_{2\eta} + j\omega)/(b_{2\eta} + j\omega)) \tag{15}$$

where $c_\eta = 1/(\kappa_{1\eta}\kappa_{2\eta})$, $b_{m\eta} = (\alpha_{m\eta}/\varepsilon_0) + (\sigma_{m\eta}/\kappa_{m\eta}\varepsilon_0)$ and $a_{m\eta} = a_{m\eta}/\varepsilon_0,\ m = 1, 2$.

Substituting (15) into (11), we can obtain

$$j\omega D_x = c_y((a_{1y} + j\omega)/(b_{1y} + j\omega))((a_{2y} + j\omega)/(b_{2y} + j\omega))\partial_y H_z - c_z((a_{1z} + j\omega)/(b_{1z} + j\omega))((a_{2z} + j\omega)/b_{2z} + j\omega)\partial_z H_y \tag{16}$$

Introducing four auxiliary variables Φ_{xy}^n, Φ_{xz}^n, Ψ_{xy}^n, and Ψ_{xz}^n.

$$\Phi_{xy} = c_y(1/(b_{1y} + j\omega))\partial H_z/\partial y \tag{17}$$

$$\Psi_{xy} = ((a_{2y} + j\omega)/(b_{2y} + j\omega))\Phi_{xy} \tag{18}$$

$$\Phi_{xz} = c_z(1/(b_{1z} + j\omega))\partial H_y/\partial z \tag{19}$$

$$\Psi_{xz} = ((a_{2z} + j\omega)/(b_{2z} + j\omega))\Phi_{xz} \tag{20}$$

Simplifying (16), we can obtain

$$j\omega D_x = (a_{1y} + j\omega)\Psi_{xy} - (a_{1z} + j\omega)\Psi_{xz} \tag{21}$$

Discretizing (17)–(21) at the time step $n + 1/2$, we obtain

$$\Phi_{xy}^{n+1}\left(i+\frac{1}{2},j,k\right) = g_{y1}(j)\cdot\Phi_{xy}^{n}\left(i+\frac{1}{2},j,k\right) + g_{y2}(j)\cdot\left(\left(H_z^{n+1/2}\left(i+\frac{1}{2},j+\frac{1}{2},k\right) - H_z^{n+1/2}\left(i+\frac{1}{2},j-\frac{1}{2},k\right)\right)\Big/\Delta y\right) \tag{22}$$

$$\Psi_{xy}^{n+1}\left(i+\frac{1}{2},j,k\right) = f_{y1}(j)\cdot\Psi_{xy}^{n}\left(i+\frac{1}{2},j,k\right) + f_{y2}(j)\cdot\Phi_{xy}^{n+1}\left(i+\frac{1}{2},j,k\right) - f_{y3}(j)\cdot\Phi_{xy}^{n}\left(i+\frac{1}{2},j,k\right) \tag{23}$$

$$\Phi_{xz}^{n+1}\left(i+\frac{1}{2},j,k\right) = g_{z1}(k)\cdot\Phi_{xz}^{n}\left(i+\frac{1}{2},j,k\right) + g_{z2}(k)\cdot\left(\left(H_y^{n+1/2}\left(i+\frac{1}{2},j,k+\frac{1}{2}\right) - H_y^{n+1/2}\left(i+\frac{1}{2},j,k-\frac{1}{2}\right)\right)\Big/\Delta z\right) \tag{24}$$

$$\Psi_{xz}^{n+1}\left(i+\frac{1}{2},j,k\right) = f_{z1}(k)\cdot\Psi_{xz}^{n}\left(i+\frac{1}{2},j,k\right) + f_{z2}(k)\cdot\Phi_{xz}^{n+1}\left(i+\frac{1}{2},j,k\right) - f_{z3}(k)\cdot\Phi_{xz}^{n}\left(i+\frac{1}{2},j,k\right) \tag{25}$$

$$D_x^{n+1} = D_x^n + w_{y1}\Psi_{xy}^{n+1} - w_{y2}\Psi_{xy}^n - (w_{z1}\Psi_{xz}^{n+1} - w_{z2}\Psi_{xz}^n) \tag{26}$$

where $f_{y1} = (2 - b_{2y}\Delta t)/(2 + b_{2y}\Delta t)$, $f_{y2} = (2 + a_{2y}\Delta t)/(2 + b_{2y}\Delta t)$, $f_{y3} = (2 - a_{2y}\Delta t)/(2 + b_{2y}\Delta t)$, $g_{y1} = (2 - b_{1y}\Delta t)/(2 + b_{1y}\Delta t)$, $g_{y2} = (2c_y\Delta t)/(2 + b_{1y}\Delta t)$, $w_{y1} = c_0(1 + a_{1y}\Delta t/2)$, $w_{y2} = c_0(1 - a_{1y}\Delta t/2)$, $f_{z1} = (2 - b_{2z}\Delta t)/(2 + b_{2z}\Delta t)$, $f_{z2} = (2 + a_{2z}\Delta t)/(2 + b_{2z}\Delta t)$, $f_{z3} = (2 - a_{2z}\Delta t)/(2 + b_{2z}\Delta t)$, $g_{z1} = (2 - b_{1z}\Delta t)/(2 + b_{1z}\Delta t)$, $g_{z2} = (2c_z\Delta t)/(2 + b_{1z}\Delta t)$, $w_{z1} = c_0(1 + a_{1z}\Delta t/2)$, $w_{z2} = c_0(1 - a_{1z}\Delta t/2)$ where Δy and Δz are space step. All coefficients, which are calculated once prior to the field computation, are stored in one-dimensional vector arrays, auxiliary variables and field components are evaluated at the corresponding Yee grid position. In order to make an equal comparison between the proposed higher-order formulations and reference [8], we assume that the proposed higher-order formulations and reference [8] use the temporary storage variables technique, without exception. Note that storage of $\Phi_{xy}|_{i+1/2,j,k}^n$, $\Phi_{xz}|_{i+1/2,j,k}^n$, $\Psi_{xy}|_{i+1/2,j,k}^n$ and $\Psi_{xz}|_{i+1/2,j,k}^n$ can be avoided by means

of temporary storage variables [9]. Consequently, the FDTD implementation of (1) by using (18)–(22) requires only the storage of four auxiliary variables per field component per cell in the SC-PML region.

The above formulations are applied to the corner PML regions and the edge PML regions which run parallel with x direction (i.e. both $S_y \neq 1$ and $S_z \neq 1$ are chosen as in (10)). In the faces of the PML regions which are normal to y direction and the edge PML regions which run parallel with z direction (i.e. $S_y \neq 1$ as in (10) and $S_z = 1$ are chosen), D_x can be obtained by using only two auxiliary variables, Φ_{xy} and Ψ_{xy}. Likewise, in the faces of the PML regions which are normal to z direction and the edge PML regions which run parallel with y direction (i.e. $S_y = 1$ and $S_z \neq 1$ as in (10) are chosen), E_x can be obtained by using only two auxiliary variables, Φ_{xz} and Ψ_{xz}. For the face PML regions which are perpendicular to x direction (i.e. both $S_y = 1$ and $S_z = 1$ are chosen), no auxiliary variable is required. Similar formulations can be obtained for other field components of D (and B).

In order to perform a comparison of the proposed formulations and the formulations in [8] in terms of memory, we assume a vacuum 3D FDTD computational domain with $L \times M \times N$ cells. A W-cell thick PML is used to truncate the FDTD computational domain on all directions. To obtain E_x, the proposed formulations require $4 \times (L + 2W) \times W \times W \times 4 = 32W^3 + 16LW^2$ auxiliary variables in the corner PML regions and the edge PML regions which run parallel with x direction (i.e. both $S_y \neq 1$ and $S_z \neq 1$ are chosen, and P_{xy}, Q_{xy}, P_{xz} and Q_{xz} are required.), $2 \times (L + 2W) \times W \times N \times 2 = 8NW^2 + 4LNW$ auxiliary variables in the faces of the PML regions which are normal to y direction and the edge PML regions which run parallel with z direction (i.e. $S_y \neq 1$ and $S_z = 1$ are chosen, and P_{xy} and Q_{xy} are required.), $2 \times (L + 2W) \times M \times W \times 2 = 8MW^2 + 4LMW$ auxiliary variables in the faces of the PML regions which are normal to z direction and the edge PML regions which run parallel with y direction (i.e. $S_y = 1$ and $S_z \neq 1$ are chosen, and P_{xz} and Q_{xz} are required.), and 0 auxiliary variables in the face PML regions which are perpendicular to x direction (i.e. both $S_y = 1$ and $S_z = 1$ are chosen). By using the same means, auxiliary variables to obtain E_y, E_z, H_x, H_y, and H_z can be computed. In consequence, the proposed formulations require $192W^3 + 64(LW^2 + MW^2 + NW^2) + 16(LMW + LNW + MNW)$ auxiliary variables in total to obtain 6 field components. However, the formulations in [8] require $288W^3 + 112(LW^2 + MW^2 + NW^2) + 32(LMW + LNW + MNW)$ auxiliary variables. Therefore, the proposed formulations require $96W^3 + 48(LW^2 + MW^2 + NW^2) + 16(LMW + LNW + MNW)$ auxiliary variables less than the formulations in [8] in the PML regions.

As compared with [8], savings in the computation time can be achieved by the proposed algorithm due to the fact that the decrease of the auxiliary variables leads to the reduction of the calculation steps. The more the cells in the PML regions, the more the savings in memory and computation time.

It is obvious that, because of no polynomial expansion, the proposed implementation of the higher-order PML is easier than that in [8] to be extended to the case with more than two poles.

3. Numerical result

Two numerical tests are provided to validate the proposed method. In the first test, to validate the proposed methods, we implement the FDTD simulation for an inhomogeneous, dispersive, and conductive soil half-space problem in a highly elongated FDTD grid. It is shown in Fig. 1. The dielectric constant of soil is specified as the second-order Debye model with an added conductivity term $\epsilon_r(\omega) = \epsilon_\infty + \sigma/j\omega + \epsilon_0 \sum_{p=1}^{2} A_p/(1 + j\omega\tau_p)$, where $\varepsilon_\infty = 4.15$ is the infinite frequency permittivity, $A_1 = 1.8$ and $A_2 = 0.6$ are the pole amplitudes, $\tau_1 = 3.79$ ns and $\tau_2 = 0.151$ ns are the relaxation time and $\sigma = 1.11$ ms/m is the conductivity [10]. The half-space occupies 50% of the vertical height of the horizontally elongated simulation region. The simulation is done with a $126 \times 46 \times 26$ grid including 10-cell-thick PML layers on all sides with the space steps $\Delta x = \Delta y = \Delta z = 0.05$ m and the time step $\Delta t = 77$ ps. Assuming that the origin is at a corner of FDTD grid, a vertically polarized point source located at (13,13,13) (just above the soil) is excited by a differentiated Gaussian pulse with a half pulse bandwidth = 1155 ps. Within the PML, σ_η and κ_η are scaled using a fourth-order polynomial scaling [11] and α_η is a constant. The relative reflection error (in decibel) versus time is computed at an observation point located at (113,33,12) (at the opposing corner of FDTD grid from the source and just below the interface of free space and soil) by using

$$R_{dB}(t) = 20\log_{10}(|E_z(t) - E_{z\text{ref}}(t)|/E_{z\text{ref_max}}) \tag{27}$$

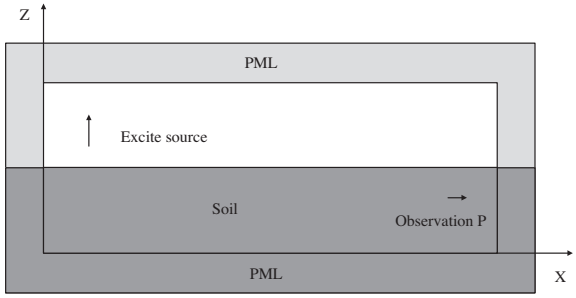


Fig. 1. An inhomogeneous, dispersive, and conductive soil half-space problem in a highly elongated FDTD grid.

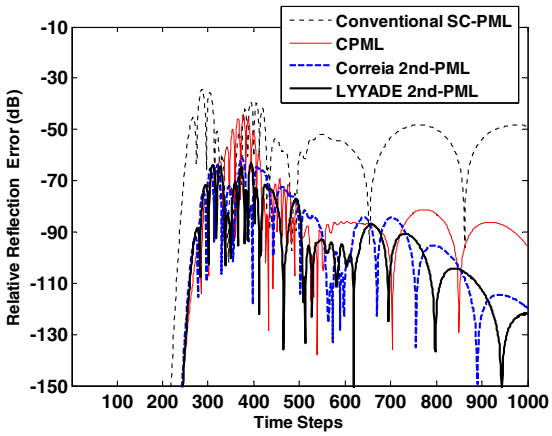


Fig. 2. Relative reflection error versus time steps: Conventional SC-PML (PML layers = 10), CPML (PML layers = 10), Correia 2nd-PML (PML layers = 10), LYYADE 2nd-PML (PML layers = 10).

Table 1
Memory requirements (bytes) for the different PML implementations.

	PML layers = X
Correia 2nd-PML	49,506 K (X = 10)
LYYADE 2nd-PML	30,604 K (X = 10)
CPML	24,084 K (X = 10)
SC-PML	24,084 K (X = 10)

Table 2
Computational time (s) for the different PML implementations.

	Time steps = 1000, (PML layers = X)
Correia 2nd-PML	667.60 (X = 10)
LYYADE 2nd-PML	412.70 (X = 10)
CPML	285.84 (X = 10)
SC-PML	303.62 (X = 10)

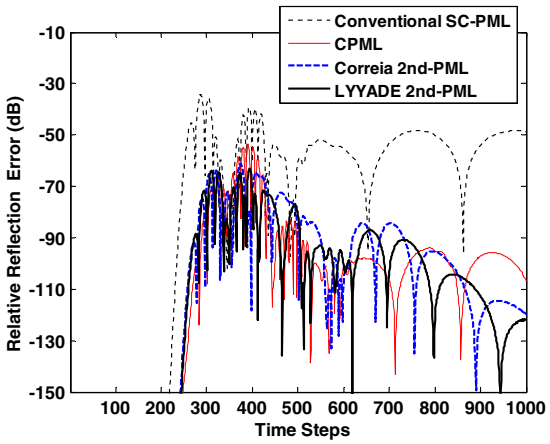


Fig. 3. Relative reflection error versus time steps: Conventional SC-PML (PML layers = 10), CPML (PML layers = 13), Correia 2nd-PML (PML layers = 10), LYYADE 2nd-PML (PML layers = 10).

where $E_z(t)$ represents the time-dependent discrete electric field of the observation point, $E_{\text{zref}}(t)$ is a reference solution based on a large $206 \times 126 \times 106$-cell grid, and $E_{\text{zref_max}}$ represents the maximum value of the reference solution over the full-time simulation. The relative reflection error is first computed over 1000 time iterations. This same example is repeated with

Table 3
Memory requirements (bytes) for the different PML implementations.

	PML layers = X
Correia 2nd-PML	49,506 K (X = 10)
LYYADE 2nd-PML	30,604 K (X = 10)
CPML	35.436 K (X = 13)
SC-PML	24,084 K (X = 10)

Table 4
Computational time (s) for the different PML implementations.

	Time steps = 1000, (PML layers = X)
Correia 2nd-PML	667.60 (X = 10)
LYYADE 2nd-PML	412.70 (X = 10)
CPML	434.73 (X = 13)
SC-PML	303.62 (X = 10)

Table 5
Memory requirements (bytes) for the different PML implementations.

	PML layers = X
Correia 2nd-PML	2056 K (X = 10)
LYYADE 2nd-PML	1376 K (X = 10)
CPML	1160 K (X = 10)
SC-PML	1252 K (X = 10)

Table 6
Computational time (s) for the different PML implementations.

	Time steps = 20,000, (PML layers = X)
Correia 2nd-PML	132.47 (X = 10)
LYYADE 2nd-PML	88.31 (X = 10)
CPML	66.54 (X = 10)
SC-PML	66.75 (X = 10)

Table 7
Memory requirements (bytes) for the different PML implementations.

	PML layers = X
Correia 2nd-PML	2056 K (X = 10)
LYYADE 2nd-PML	1376 K (X = 10)
CPML	1316 K (X = 13)
SC-PML	1252 K (X = 10)

SC-PML ($\alpha_\eta = 0$, $\kappa_{\max} = 11$, and $\sigma_{\max} = 0.18$ S/m) and the convolutional PML (CPML) [4] ($\alpha_\eta = 0.0015$, $\kappa_{\max} = 7$, and $\sigma_{\max} = 0.24$S/m). For the second-order PML including the formulations in [8] and the proposed formulations, the following parameters are chosen:

$\kappa_{1\eta} = 1$, $\alpha_{1\eta} = 0$, $\sigma_{1\eta\text{opt}} = 0.175/150\pi\Delta x$, $\sigma_{1\eta} = \sigma_{1\eta\text{opt}}\rho^4$, $\kappa_{2\eta\text{opt}} = 9$, $\kappa_{2\eta} = 1 + \kappa_{2\eta\text{opt}}\rho^2$, $\alpha_{2\eta} = 0.0023 + \sigma_{1\eta}$, $\sigma_{2\eta\text{opt}} = 4/150\pi\Delta x$, and $\sigma_{2\eta} = \sigma_{2\eta\text{opt}}\rho^2$, where ρ is zero at the interface of the PML and the FDTD computational domains and 1 at the end. In all computations of this paper, σ and κ are evaluated as the average value in the cell around the index location [1]. These optimum parameters are chosen empirically to obtain the lowest reflection. The difference of the optimum parameters of LYYADE PML and CPML results from different scheme (i.e., the coefficients of (22)–(26) are different from the counterpart of CPML).

These results are illustrated from Fig. 2. It is shown in Fig. 2 that the LYYADE PML and the proposed PML in [8] are similar and both have improvement near 400 time steps and much lower reflection error for the late-time region, whereas SC-PML has a comparatively high reflection error over the entire time simulation due to the oblique incidence of the waves and low-frequency evanescent fields that are interacting with the PML interface. Tables 1 and 2 are using different perfectly matched layer algorithm procedures which occupy of memory and different time steps occupy computational time, respectively.

Table 8
Computational time (s) for the different PML implementations.

	Time steps = 20,000, (PML layers = X)
Correia 2nd-PML	132.47 (X = 10)
LYYADE 2nd-PML	88.31 (X = 10)
CPML	80.48 (X = 13)
SC-PML	66.75 (X = 10)

Obviously, when FDTD computational domain is invariant, saving memory will increase with the increase of PML layers and saving time will increase with the increase of time steps.

In order to demonstrate the absorbing properties of the high order PML as compared to more conventional PML implementations, the number of PML of the CPML is increased to 13-cell-thick. It is clearly shown in Fig. 3 that the CPML (PML = 13) hold the comparatively good absorbing performance compared with the CPML (PML = 10), especially much lower reflection error for the late-time region. Whereas, it can be shown in Table 3 that the requirement of memory of the CPML has increased from 24,086 K (PML = 10) to 35,436 K (PML = 13) which is more than 30,604 K of the LYYADE PML (PML = 10). Besides, it is shown in Table 8 the LYYADE PML (PML = 10) requires less computational time compared with the CPML (PML = 13).

In the second test, to simplify the problem, but without loss of generality, we model a 2D TE-polarized electromagnetic wave interaction with an infinitely long perfectly electric conductor (PEC) sheet with the finite width to validate the proposed formulations. Fig. 4 shows the FDTD grid geometry used in this simulation. The space is discretized with the FDTD lattice with $\Delta x = \Delta y = 1$ mm and time step is $\Delta t = 1.1785$ ps. The FDTD computational domains consist of a 100-cell wide PEC sheet surrounded by free space. 10-cell thick PML layers terminate the grid and are placed only 3 cells away from the PEC sheet in all directions.

A y-polarized line electric current source, infinitely long in the z direction, is placed at the center and excited with a differentiated Gaussian pulse given by

$$J_y(t) = -2(t - t_0)/t_w \exp[-((t - t_0)/t_w)^2] \tag{28}$$

where $t_w = 26.53$ ps and $t_0 = 4t_w$. The y-component of the electric field is measured at the point P, where we expect very strong evanescent waves to appear. The relative reflection error (in dB) versus time is computed at the observation point P by using

$$R_{dB}(t) = 20\log_{10}((|E_y^R(t) - E_y^T(t)|)/|E_{y_\max}^R|) \tag{29}$$

where $E_y^T(t)$ represents the time-dependent discrete electric field of the observation point , $E_y^R(t)$ is a reference solution based on a large 1, 200 $\times$ 1, 200-cell grid, and $E_{y_\max}^R$ represents the maximum value of the reference solution over the full time simulation. The reference grid is sufficiently large such that there are no reflections from its outer boundaries during 1500 time steps which are well past the steady-state response. The same example is repeated with the conventional SC-PML (which is equivalent to the CPML with $\alpha_\eta = 0$), the CPML and the 2nd-order PML in [8]. Within the conventional SC-PML and the CPML, κ_η and σ_η are scaled by using a fourth-order polynomial scaling ($m = 4$) and α_η is a constant, as in [4]. The σ_{opt} is chosen as

$$\sigma_{\text{opt}} = (m + 1)/150\pi\Delta x \tag{30}$$

For the conventional SC-PML, $\kappa_{\max} = 9$ and $\sigma_{\max} = 0.5\sigma_{\text{opt}}$ are chosen. In the CPML simulation,$\kappa_{\max} = 9$, $\sigma_{\max} = 0.9\sigma_{\text{opt}}$ and $\alpha_\eta = 0.06$ are chosen. For the 2nd-order PML including the formulations in [8] and the proposed formulations, the following parameters are chosen: $\kappa_{1\eta} = 1$, $\alpha_{1\eta} = 0$, $\sigma_{1\eta\text{opt}} = 0.075/150\pi\Delta x$, $\sigma_{1\eta} = \sigma_{1\eta\text{opt}}\rho^4$, $\kappa_{2\eta\text{opt}} = 8$, $\kappa_{2\eta} = 1 + \kappa_{2\eta\text{opt}}\rho^2$, $\alpha_{2\eta} = 0.09 + \sigma_{1\eta}$, $\sigma_{2\eta\text{opt}} = 4/150\pi\Delta x$, and $\sigma_{2\eta} = \sigma_{2\eta\text{opt}}\rho^2$. These optimum parameters are chosen empirically to obtain the lowest reflection (See Table 4.).

The results are illustrated in Fig. 5. The maximum relative errors of the conventional SC-PML, the CPML, the second-order PML in [8] and LYYADE PML are −49 dB, −75 dB, −90 dB and −90 dB, respectively. It can be concluded from figures that the

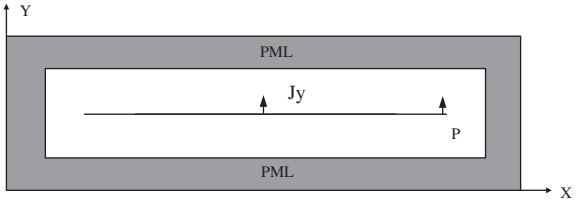


Fig. 4. Shows the FDTD grid geometry in this simulation.

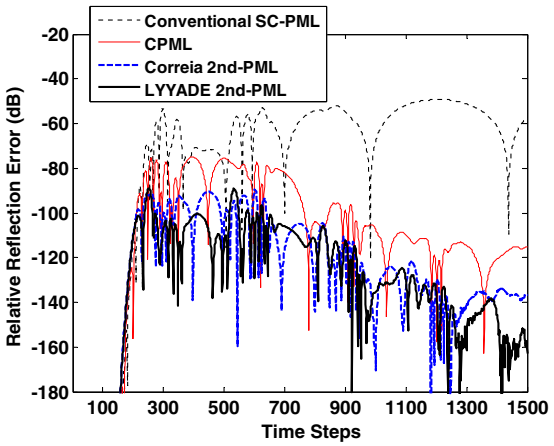


Fig. 5. Relative reflection error versus time steps: Conventional SC-PML (PML layers = 10), CPML (PML layers = 10), Correia 2nd-PML (PML layers = 10), LYYADE 2nd-PML (PML layers = 10).

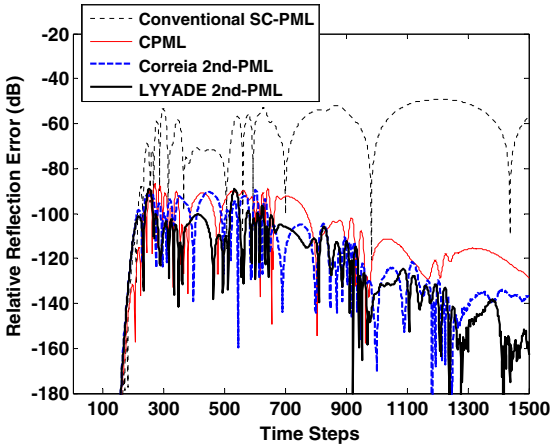


Fig. 6. Relative reflection error versus time steps: Conventional SC-PML (PML layers = 10), CPML (PML layers = 13), Correia 2nd-PML (PML layers = 10), LYYADE 2nd-PML (PML layers = 10).

absorbing performance of the proposed 2nd-order formulations is similar to that in [8] and has 15 dB and 41 dB improvement in terms of the maximum relative error as compared with the CPML and the SC-PML. Tables 5 and 6 are as the same with first test by using different perfectly matched layer algorithm procedures which occupy of memory and different time steps occupy computational time, respectively. Obviously, when FDTD computational domain is invariant, saving memory will increase with the increase of PML layers and saving time will increase with the increase of time steps.

In order to demonstrate the absorbing properties of the high order PML as compared to more conventional PML implementations, the number of PML of the CPML is increased to 13-cell-thick. It is clearly shown in Fig. 6 that the CPML (PML = 13) hold the comparatively good absorbing performance compared with the CPML (PML = 10). And then, the maximum relative error of the CPML (PML = 13) declines to −90 dB and has 15 dB improvement compared with −75 dB of the CPML (PML = 10). Whereas, it can be shown in Table 7 that the requirement of memory of the CPML has increased from 1160 K (PML = 10) to 1316 K (PML = 13). Besides, it is shown in Table 8 that the CPML (PML = 13) requires more computational time than the CPML (PML = 10). It can be concluded in Fig. 6 that the maximum relative error of the CPML (PML = 13) is similar to that of the LYYADE PML (PML = 10), whereas the LYYADE PML has lower reflection error for the late-time region than the CPML.

4. Conclusions

An efficient and unsplit-field implementation of the higher-order PML based on the SC-PML and the auxiliary differential equation method has been presented. It has been shown in the numerical tests that this novel high-order PML implementation yields the same results as the initial and more computationally demanding implementation in [7,8].

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References

[1] J.P. Bérenger, A perfectly matched layer for the absorption of electromagnetic waves, J. Comput. Phys. 114 (1994) 185–200.
[2] W.C. Chew, W.H. Weedon, A 3-D perfectly matched medium from modified Maxwell's equations with stretched coordinates, Microw. Opt. Technol. Lett. 7 (1994) 599–603.
[3] M. Kuzuoglu, R. Mittra, Frequency dependence of the constitutive parameters of causal perfectly matched anisotropic absorbers, IEEE Microw. Guided Wave Lett. 6 (1996) 447–449.
[4] J.A. Roden, S.D. Gedney, Convolution PML (CPML): an efficient FDTD implementation of the CFS-PML for arbitrary media, Microw. Opt. Technol. Lett. 27 (2000) 334–339.
[5] E. Bécache, P.G. Petropoulus, S.D. Gedney, On the long-time behavior of unsplit perfectly matched layers, IEEE Trans. Anten. Propag. 52 (2004) 1335–1342.
[6] J.P. Bérenger, Numerical reflection from FDTD-PMLs: a comparison of the split PML with the unsplit CFS PML, IEEE Trans. Anten. Propag. 50 (2002) 258–265.
[7] D. Correia, J.M. Jin, Performance of regular PML, CFS-PML, and second-order PML for waveguide problems, Microw. Opt. Technol. Lett. 48 (2006) 2121–2126.
[8] D. Correia, J.M. Jin, On the development of a higher-order PML, IEEE Trans. Anten. Propag. 53 (2005) 4157–4163.
[9] D.F. Kelley, R.J. Luebbers, Piecewise linear recursive convolution for dispersive media using FDTD, IEEE Trans. Anten. Propag. 44 (1996) 792–797.
[10] F.L. Teixeira, W.C. Chew, M. Straka, M.L. Oristaglio, T. Wang, Finite-difference time-domain simulation of ground penetrating radar on dispersive, inhomogeneous, and conductive soils, IEEE Trans. Geosci. Remote Sens. 36 (1998) 1928–1937.
[11] S.D. Gedney, The perfectly matched layer absorbing medium, in: A. Taflove (Ed.), Advance in Computational Electrodynamics: The Finite Difference Time Domain, Artech House, Boston, MA, 1998, pp. 263–340.
[12] D.M. Sullivan, Frequency-dependent FDTD methods using Z transforms, IEEE Trans. Anten. Propag. 40 (1992) 1223–1230.